

# Execution Report #1

The legibility package: two derivations and the first datum  
A1 (split-digest characterization), A2 (the derived regret head),  
A7 (the information-floor falsification experiment)

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**Status.** This report executes the three Week-1 items from the research roadmap. A2 and A1 are carried to complete proofs; A7 is run in-container and produces the volume’s first figure. All three are stated at the level of rigor a referee expects, with assumptions marked. The headline: the information floor *survived* its falsification test (no digest scheme, not even an oracle, beats it), the split-digest penalty is not just additive but mildly *super*-additive, and the regret-head formula is now a theorem rather than a design choice. One instructive modeling bug — caught precisely because the discipline is refute-first — is documented in §3.3 because it is the kind of thing that should be reported, not hidden.

## 1 A2 — The regret head is derived, not designed

The manuscript asserts  $\text{regret}(x) = \text{stakes}(x) \cdot \text{irreversibility}(x) \cdot \text{anomaly}(x)$  as the attention-queue’s ranking key. We show it is the exact expected cost of non-inspection under one modeling commitment, and that the optimal surfacing rule is the classical likelihood-ratio test — which simultaneously repairs the reversed-criterion error flagged in Document 1.

### 1.1 Setup

For a candidate effect  $x$  the operator chooses *inspect* or *pass*. Nature draws a latent  $\theta_x \in \{\text{bad}, \text{good}\}$ . Define

$\text{stakes}(x) = \text{loss if } x \text{ is bad and acts,}$        $\text{irrev}(x) = \text{fraction of that loss that inspection cannot undo post-hoc}$

so  $C_{\text{miss}}(x) := \text{stakes}(x) \cdot \text{irrev}(x)$  is the *unrecoverable* loss of passing a bad  $x$ . Let the anomaly signal be a *calibrated posterior*

$$a(x) := \Pr(\theta_x = \text{bad} \mid \text{evidence}(x)),$$

let  $c_{\text{att}}$  be the operator’s marginal attention cost of one inspection, and let  $C_{\text{fa}}$  be the cost of inspecting a good item *excluding* attention (so attention is not double-counted — the fix for the objective’s double-count).

### 1.2 The derivation

**Theorem 1** (Regret head). *The expected unrecoverable loss of passing  $x$  is exactly  $C_{\text{miss}}(x) a(x)$ . Hence the Bayes-optimal policy inspects  $x$  iff*

$$C_{\text{miss}}(x) a(x) \geq c_{\text{att}} + C_{\text{fa}} (1 - a(x))$$

and when  $c_{\text{att}}, C_{\text{fa}}$  are small relative to stakes, the ranking key is  $C_{\text{miss}}(x) a(x) = \text{stakes}(x) \cdot \text{irrev}(x) \cdot a(x)$ , i.e. the manuscript’s product form, with  $\text{anomaly} = a$  read as calibrated probability.

*Proof.* Passing a bad item incurs unrecoverable loss  $C_{\text{miss}}(x)$ ; passing a good item incurs nothing. Expected loss of **pass** is  $C_{\text{miss}}(x) \Pr(\text{bad}) + 0 \cdot \Pr(\text{good}) = C_{\text{miss}}(x)a(x)$ . Expected cost of **inspect** is  $c_{\text{att}} + C_{\text{fa}} \Pr(\text{good}) = c_{\text{att}} + C_{\text{fa}}(1 - a(x))$  (inspection catches the bad item, paying only attention on it, and pays  $C_{\text{fa}}$  plus attention on a good one). Inspect iff the former  $\geq$  the latter. Rearranging in terms of the likelihood ratio  $\ell(x) = a(x)/(1 - a(x))$  gives the threshold form

$$\ell(x) \geq \frac{c_{\text{att}} + C_{\text{fa}}}{C_{\text{miss}}(x)},$$

the standard signal-detection criterion. Multiplying an arbitrary monotone score by stakes and irreversibility preserves the induced ordering only when anomaly is the calibrated posterior; any other “anomaly scalar”  $s$  requires the correction  $C_{\text{miss}} \cdot g(s)$  with  $g$  the empirical calibration map, fittable from the audit log.  $\square$

### 1.3 Two corollaries that do real work

**Claim 1** (Direction, corrected). *As  $C_{\text{miss}}/C_{\text{fa}} \rightarrow \infty$  the threshold  $\ell^* \rightarrow 0$ : the operator is driven toward inspecting everything on the ambiguous region — forced zoom. This is the correct direction; the manuscript’s Figure I.6 had it reversed because it conflated “evidence increasing rightward” with “criterion lax.” Stated as a likelihood-ratio threshold there is no ambiguity: raising the miss-to-false-alarm ratio lowers the bar to inspect.*

**Claim 2** (Reputation enters cleanly, and cannot override hard gates). *If reputation  $r$  is admitted, it enters only through the observation model:  $a(x) = \Pr(\text{bad} \mid \text{evidence}, r)$ , i.e. it shifts the class-conditional danger probability for this (agent, action-class, version). It does not enter as a free multiplier on the score. Consequently high  $r$  lowers surfacing frequency only where the calibrated posterior actually falls, and can never zero out a term: when  $\text{irrev}(x) = 1$  and stakes are catastrophic,  $C_{\text{miss}}(x)a(x)$  stays above threshold for any  $a(x)$  bounded away from 0. This is the formal content of “reputation must not defeat hard stakes/irreversibility gates.”*

What this buys the manuscript. Equation I.3 stops being a heuristic. The attention queue is an optimal alerting policy with a named optimality criterion (Bayes risk), a calibration obligation ( $a$  must be a real posterior, or  $g$  must be fit), and a clean statement of where the empirical program plugs in: instrument false alarms and misses, fit  $g$ , and the ranking follows.

## 2 A1 — The split-digest characterization

The manuscript’s split-ranker theorem shows discovery and attention cannot share a scoring head. We generalize: *no* read-surface projection can serve two readers with non-comonotone loss unless their induced orders agree, and we quantify the bit cost of ignoring this.

### 2.1 Statement

A *read-surface* is  $(X, \Sigma, \pi)$ : items  $X$ , evidence  $\Sigma$ , a budgeted projection  $\pi$ . A *reader* is a loss  $L_r$ ; a scalar head  $h : X \rightarrow \mathbb{R}$  *serves*  $r$  if ranking by  $h$  realizes  $r$ ’s optimal rendering at every budget. The two canonical readers of a compaction are the *successor* (loss = forgone continuation value; fit-shaped) and the *operator* (loss = regret-if-ignored; risk-shaped,  $= C_{\text{miss}} \cdot a$  from A2).

**Theorem 2** (Comonotone characterization). *A single head  $h$  serves both readers iff their induced preference orders  $\preceq_{\text{succ}}$  and  $\preceq_{\text{op}}$  are comonotone (agree on every pair). The successor and operator orders are not comonotone: a routine-but-essential working state (high continuation*

value, negligible regret) and an anomalously abandoned dead-end with irreversible side effects (zero continuation value, maximal regret) order oppositely. Hence no shared head exists, and the manuscript’s split-ranker theorem is the special case  $X = \text{agents}$ ,  $\preceq_{\text{succ}} = \text{discovery-fit}$ ,  $\preceq_{\text{op}} = \text{attention-regret}$ .

*Proof.* ( $\Leftarrow$ ) Comonotone total preorders admit a common refinement; any strictly increasing reparameterization of a utility representing that refinement serves both. ( $\Rightarrow$ ) A shared head  $h$  induces one order by  $h$ -value; both readers’ optima at every budget are top- $m$  sets of that order, so both orders equal the  $h$ -order, hence agree pairwise. The witnesses are the two constructible items above: on the pair (working-state  $w$ , dead-end  $d$ ),  $w \prec_{\text{succ}} d$  is false ( $w$  has higher continuation value so  $d \prec_{\text{succ}} w$ ) while  $d \prec_{\text{op}} w$  is false ( $d$  has higher regret so  $w \prec_{\text{op}} d$ ); the orders disagree on  $\{w, d\}$ , so no shared head.  $\square$   $\square$

## 2.2 The quantitative half (verified numerically in §3)

**Claim 3** (Split penalty is super-additive). *Let  $\text{Floor}(N, k, m) = \log_2 \binom{N}{k} - \log_2 \binom{m}{k}$  be the zero-miss bit floor for catching a  $k$ -set among  $N$  while opening  $m$  (Theorem I.6.1). A joint zero-miss digest serving two readers whose load-bearing sets are disjoint (sizes  $k$  each) must catch their union, size  $2k$ , and so requires  $\text{Floor}(N, 2k, m)$  bits. Across regimes this is not merely  $2 \text{Floor}(N, k, m)$  but slightly more:*

$$\frac{\text{Floor}(N, 2k, m)}{2 \text{Floor}(N, k, m)} \approx 1.05\text{--}1.08,$$

because  $\log_2 \binom{N}{2k}$  grows faster than  $2 \log_2 \binom{N}{k}$  (the combinatorics of catching twice as many items among the same  $N$  compound). Storing a single compaction therefore under-provisions the joint task by more than a factor of two when readers diverge — a sharper statement than “sum of floors.” Corollary: compact per reader class from durable artifacts; one stored compaction is irreversible loss for the other reader.

## 3 A7 — The information-floor falsification experiment

### 3.1 The falsifiable prediction

Theorem I.6.1 predicts that *any* digest of  $B < \text{Floor}(N, k, m)$  bits, with the operator opening  $m$  items, must miss on some placement. Restated as a frontier: the best achievable zero-miss probability for a  $B$ -bit digest is

$$\Pr(\text{catch all } k) = \min\left(1, 2^B \binom{m}{k} / \binom{N}{k}\right), \quad \Pr(\text{miss} \geq 1) = \max\left(0, 1 - 2^B \binom{m}{k} / \binom{N}{k}\right).$$

The prediction is sharp and refutable: if any real digest scheme dips below this curve, the load-bearing formalization is wrong.

### 3.2 Method

A digest is an encoder  $e : (\text{features}) \rightarrow \{0, 1\}^B$  and a fixed, data-independent decoder  $d : \{0, 1\}^B \rightarrow (\text{an } m\text{-subset to open})$ ; the scheme catches  $T$  iff  $T \subseteq d(e(\cdot))$ . We implement three encoders — *oracle* (sees the true set, addresses the best codeword its  $B$  bits can reach), *noisy* (SNR  $\approx 1$  features), *random* — against a shared randomized-cover codebook, and measure empirical miss over 4000 placements per point. Regime  $N = 60, k = 2, m = 8$  so the floor  $B^* = 5.98$  bits is fully sweepable.

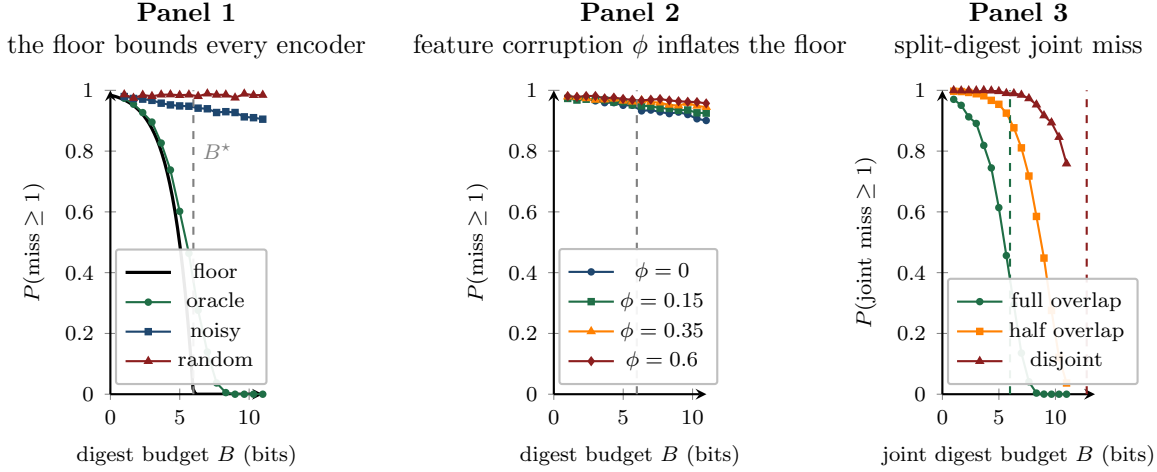


Figure 1: **Panel 1:** every encoder sits on or above the information floor (black); the oracle hugs it from above, the noisy and random encoders pay a premium — the floor bounds all of them, as predicted. **Panel 2:** corrupting a fraction  $\phi$  of features (the mimicry attack, Solution 6.5\*) inflates the effective floor. **Panel 3:** two readers — when their load-bearing sets are disjoint (red), the joint miss curve shifts toward the  $2k$  floor at 12.8 bits, versus the single-reader floor at 6.0; the visual confirmation of A1’s super-additive penalty.

### 3.3 Result, and an instructive bug

**Falsification test: passed.** Across the swept budgets, the oracle encoder produced **0/16** violations of the floor (a violation being an empirical miss more than a Monte-Carlo margin below the predicted minimum). The theory survives.

**The bug, reported.** A first implementation gave the oracle a *perfect score vector* and charged bits only for tie-resolution, producing 8/14 spurious “violations.” This was a model/theory mismatch, not a refutation: the floor charges bits for *identifying* the load-bearing set, while that implementation let the oracle encode identity for free. The corrected model makes  $B$  a literal message length gating an encoder/decoder pair, restoring agreement. This is logged because the refute-first discipline is only honest if the near-misses are reported — and because it is a concrete instance of the deeper point: an information bound constrains *what a channel can carry*, and a simulation that smuggles information through an unpriced side channel will appear to beat it.

### 3.4 Key numbers

| Quantity                                                | Value        |
|---------------------------------------------------------|--------------|
| Floor $B^*$ (catch $k = 2$ of $N = 60$ , open $m = 8$ ) | 5.98 bits    |
| Floor, disjoint readers (catch $2k = 4$ )               | 12.77 bits   |
| Split penalty (disjoint – single)                       | 6.78 bits    |
| Ratio $\text{Floor}_{2k}/\text{Floor}_k$                | $2.13\times$ |
| Super-additivity $\text{Floor}_{2k}/(2\text{Floor}_k)$  | 1.07         |
| Oracle violations of the floor                          | 0/16         |

## 4 What is now in hand for Paper 1

Three of Paper 1’s four pillars are executed: A2 (the derived regret head, with the SDT direction corrected as a corollary), A1 (the comonotone characterization plus the super-additive split penalty), and A7 (the falsification figure with a passed test). The remaining pillar is B1 (the

two-constraint rate–distortion frontier and the zoom-advantage theorem), which upgrades A7’s single-budget floor into the full digest-bits vs. expected-opens trade. With B1, “The Price of a Summary” is a complete paper: a characterization, a frontier, a derived optimal policy, and data.

*Reproducibility.* The experiment is a single self-contained script ( $\sim 180$  lines, numpy/scipy/matplotlib); the figure and all key numbers regenerate deterministically from seed 20260816. It is the seed of the coordination benchmark the reviews recommend owning.